

Task

Perform an exploratory data analysis of the `/content/sample_full.csv` dataset, focusing on learning outcomes, including visualizations with legends, and summarize the key findings.

Load Dataset

Subtask:

Load the `/content/sample_full.csv` dataset into a pandas DataFrame for initial inspection.

Reasoning: Import the pandas library and load the specified CSV file into a DataFrame, then display the first few rows for verification.

```
In [1]: import os
import pandas as pd

CSV_PATH = "content/sample_full.csv" # apni CSV ka exact naam do
assert os.path.exists(CSV_PATH), f"CSV file not found: {CSV_PATH}"

df = pd.read_csv(CSV_PATH)
print("Dataset size:", len(df))
df.head()
```

Out[1]: Dataset size: 21380

Out[1]:

	Learning_outcome	Remember	Understand	Apply	Analyze	Evaluate	Create
0	Analyze the health economic implications of e...	NaN	NaN	NaN	1.0	NaN	NaN
1	Apply research skills to operate effectively ...	NaN	NaN	1.0	NaN	NaN	NaN
2	Assess and synthesise diverse information abo...	NaN	NaN	NaN	NaN	1.0	1.0
3	Describe the general characteristics of the m...	NaN	1.0	NaN	NaN	NaN	NaN
4	Evaluate the different models of perioperativ...	NaN	NaN	NaN	NaN	1.0	NaN

```
In [2]: print('Missing values per column:')
print(df.isnull().sum())
```

Out[2]: Missing values per column:

```
Learning_outcome      0
Remember              20195
Understand            15555
Apply                 15299
Analyze               17921
Evaluate              17546
Create                17493
dtype: int64
```

Initial Data Exploration

Subtask:

Perform initial data exploration to understand the structure, data types, identify missing values, and get a summary of descriptive statistics.

Reasoning: To perform initial data exploration, I will use `.info()` to get a summary of the DataFrame's structure and data types, `.isnull().sum()` to identify missing values in each column, and `.describe()` to obtain descriptive statistics for numerical columns.

```
In [3]: print(df.info())
print("\nMissing values per column:")
print(df.isnull().sum())
print("\nDescriptive statistics:")
print(df.describe())
```

```
Out[3]: <class 'pandas.core.frame.DataFrame'>
RangeIndex: 21380 entries, 0 to 21379
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype  
---  -
 0   Learning_outcome 21380 non-null object  
 1   Remember         1185 non-null  float64
 2   Understand       5825 non-null  float64
 3   Apply           6081 non-null  float64
 4   Analyze         3459 non-null  float64
 5   Evaluate        3834 non-null  float64
 6   Create          3887 non-null  float64
dtypes: float64(6), object(1)
memory usage: 1.1+ MB
None

Missing values per column:
Learning_outcome      0
Remember              20195
Understand            15555
Apply                 15299
Analyze               17921
Evaluate              17546
Create                17493
```

dtype: int64

Descriptive statistics:

	Remember	Understand	Apply	Analyze	Evaluate	Create
count	1185.0	5825.0	6081.0	3459.0	3834.0	3887.0
mean	1.0	1.0	1.0	1.0	1.0	1.0
std	0.0	0.0	0.0	0.0	0.0	0.0
min	1.0	1.0	1.0	1.0	1.0	1.0
25%	1.0	1.0	1.0	1.0	1.0	1.0
50%	1.0	1.0	1.0	1.0	1.0	1.0
75%	1.0	1.0	1.0	1.0	1.0	1.0
max	1.0	1.0	1.0	1.0	1.0	1.0

Analyze Learning Outcomes

Subtask:

Investigate the columns related to 'Learning Outcomes' to understand their distribution and unique values, and identify any patterns or categories.

Reasoning: To investigate the 'Learning_outcome' column and cognitive domain columns, I will first get the count of unique learning outcomes and display some samples. Then, for each cognitive domain column, I will sum the non-null values to determine the number of learning outcomes associated with each domain.

```
In [4]: print(f"Number of unique learning outcomes: {df['Learning_outcome'].nunique()}")
print("\nSample unique learning outcomes:")
print(df['Learning_outcome'].unique()[:5])

cognitive_domains = ['Remember', 'Understand', 'Apply', 'Analyze', 'Evaluate', 'Create']
cognitive_domain_counts = df[cognitive_domains].count()

print("\nLearning outcomes per cognitive domain:")
print(cognitive_domain_counts)
```

```
Out[4]: Number of unique learning outcomes: 21246
```

Sample unique learning outcomes:

```
[' Analyze the health economic implications of efficient perioperative practice. '
 ' Apply research skills to operate effectively as a member of a research project team. '
 ' Assess and synthesise diverse information about information and knowledge management
technologies market and how to use implementation strategies to maximise their strengths and
minimise their weaknesses. '
 ' Describe the general characteristics of the modern X-ray system used in clinical practice,
including scientific principles, and production of the digital image. '
 ' Evaluate the different models of perioperative care that exist globally. ']
```

Learning outcomes per cognitive domain:

```
Remember      1185
```

```
Understand    5825
Apply         6081
Analyze       3459
Evaluate      3834
Create        3887
dtype: int64
```

Visualize Key Metrics

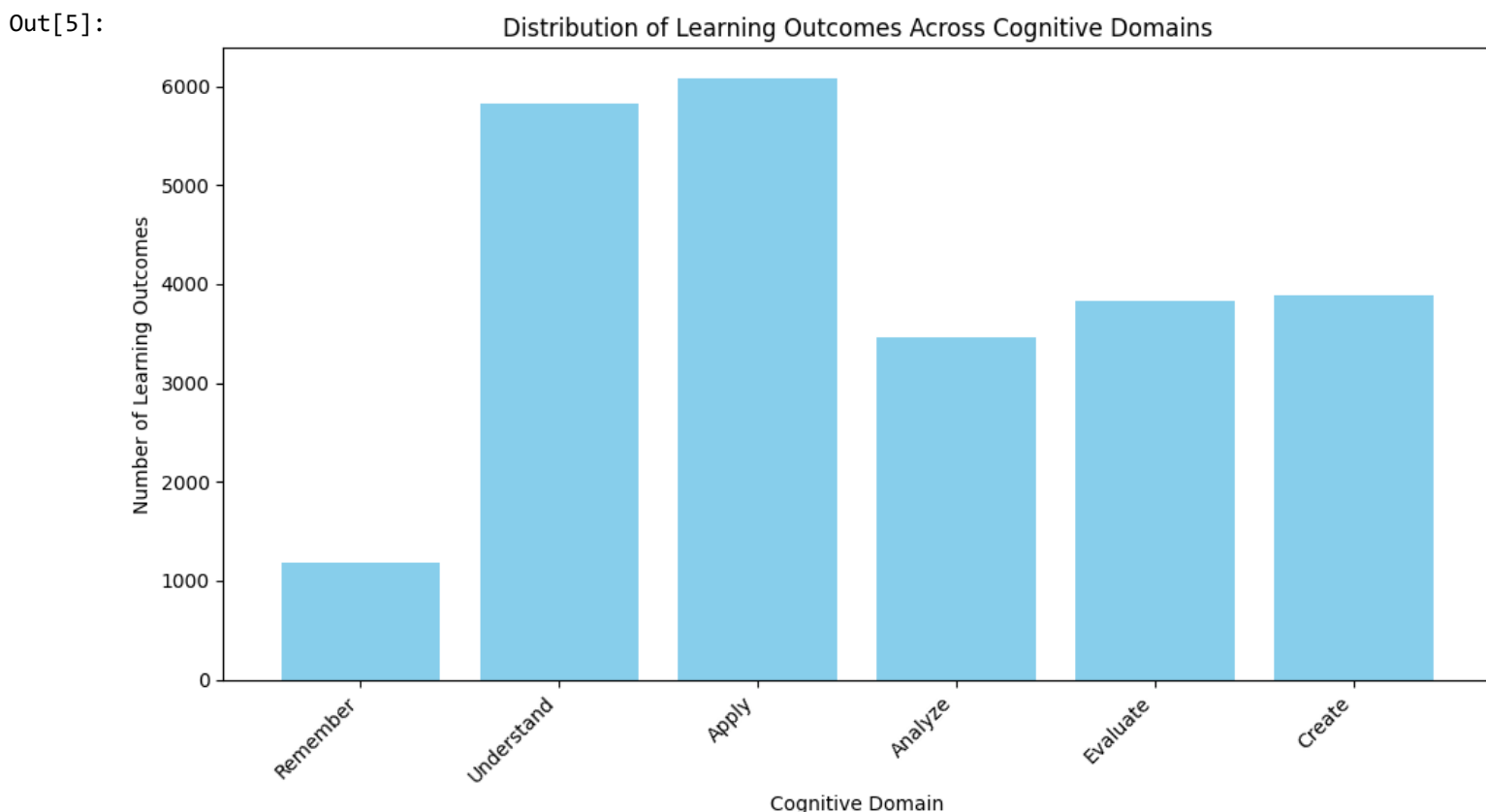
Subtask:

Generate visualizations, such as bar charts or histograms, to illustrate the distribution of learning outcomes or other relevant categorical/numerical data, including legends for clarity.

Reasoning: To visualize the distribution of learning outcomes across cognitive domains, I will create a bar chart using `matplotlib.pyplot` and label it appropriately for clarity.

```
In [5]: import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))
plt.bar(cognitive_domain_counts.index, cognitive_domain_counts.values, color='skyblue')
plt.title('Distribution of Learning Outcomes Across Cognitive Domains')
plt.xlabel('Cognitive Domain')
plt.ylabel('Number of Learning Outcomes')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Final Task

Subtask:

Summarize the key findings from the exploratory data analysis of the Moonash University learning outcomes dataset.

Summary:

Data Analysis Key Findings

- The dataset contains 21,380 entries and 7 columns, including `Learning_outcome` (object type) and six cognitive domain columns (`Remember`, `Understand`, `Apply`, `Analyze`, `Evaluate`, `Create`) which are `float64` type.
- The `Learning_outcome` column has no missing values, but all cognitive domain columns have a very high number of missing values, ranging from 15,299 (`Apply`) to 20,195 (`Remember`), indicating significant data sparsity.
- For non-null entries, the values in the cognitive domain columns are uniformly 1.0 with a standard deviation of 0.0, suggesting they act as binary indicators (1.0 or NaN) rather than continuous variables.
- There are 21,246 unique entries in the `Learning_outcome` column, indicating a high diversity of learning outcomes.
- The distribution of learning outcomes across cognitive domains shows:
 - `Remember`: 1,185
 - `Understand`: 5,825
 - `Apply`: 6,081
 - `Analyze`: 3,459
 - `Evaluate`: 3,834
 - `Create`: 3,887
 - `Apply` (6,081) and `Understand` (5,825) are the most frequently associated cognitive domains, while `Remember` (1,185) has the fewest associations.
- A bar chart visually confirmed that 'Apply' and 'Understand' domains have the highest counts of associated learning outcomes, and 'Remember' has the lowest.

Insights or Next Steps

- Given the binary nature of the cognitive domain columns and the high proportion of missing values, further investigation is needed to understand if 'NaN' implies the absence of that cognitive domain or genuinely missing data. This will influence how these columns are used for analysis.
- Due to the large number of unique `Learning_outcome` entries, natural language processing (NLP) techniques could be applied to categorize or cluster similar learning outcomes, providing deeper insights into the curriculum structure.

Bloom Taxonomy Classification using Open Source LLMs

This notebook evaluates open-source Large Language Models on the **Monash University Learning Outcomes dataset**.

Objective

Classify learning outcomes into Bloom's Taxonomy levels:

- Remember
- Understand
- Apply
- Analyze
- Evaluate
- Create

Approach

1. Load dataset
2. Convert one-hot labels to single label
3. Split dataset
4. Prompt LLM for classification
5. Evaluate predictions

Models tested:

- Mistral 7B Instruct
- Gemma 2B IT
- Llama 3 8B Instruct

◆ 1. Environment Setup

```
In [6]: %pip install -U pandas numpy scikit-learn matplotlib tqdm datasets evaluate
        %pip install -U transformers accelerate sentencepiece
        %pip install -U bitsandbytes
        %pip uninstall torch torchvision torchaudio -y
        %pip uninstall nvidia-* -y
        %pip uninstall nvidia-cuda-runtime-cu12 nvidia-cublas-cu12 nvidia-cusparse-cu12 -y
        %pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118
```

```
Out[6]: Requirement already satisfied: pandas in /usr/local/lib/python3.11/dist-packages (2.2.0)
        Collecting pandas
```

```
Downloading pandas-3.0.2-cp311-cp311-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (79 kB)
79.5/79.5 kB 2.4 MB/s eta 0:00:00
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (1.26.3)
Collecting numpy
  Downloading numpy-2.4.4-cp311-cp311-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.11/dist-packages
(1.3.0)
Collecting scikit-learn
  Downloading scikit_learn-1.8.0-cp311-cp311-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (11 kB)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.11/dist-packages (3.7.3)
Collecting matplotlib
  Downloading matplotlib-3.10.8-cp311-cp311-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (52 kB)
52.8/52.8 kB 13.3 MB/s eta 0:00:00
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (4.66.1)
Collecting tqdm
  Downloading tqdm-4.67.3-py3-none-any.whl.metadata (57 kB)
57.7/57.7 kB 15.0 MB/s eta 0:00:00
Requirement already satisfied: datasets in /usr/local/lib/python3.11/dist-packages (2.14.5)
Collecting datasets
  Downloading datasets-4.8.4-py3-none-any.whl.metadata (19 kB)
Collecting evaluate
  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-
packages (from pandas) (2.8.2)
Requirement already satisfied: scipy>=1.10.0 in /usr/local/lib/python3.11/dist-packages (from
scikit-learn) (1.11.2)
Requirement already satisfied: joblib>=1.3.0 in /usr/local/lib/python3.11/dist-packages (from
scikit-learn) (1.3.2)
Requirement already satisfied: threadpoolctl>=3.2.0 in /usr/local/lib/python3.11/dist-
packages (from scikit-learn) (3.2.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages
(from matplotlib) (1.2.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (from
matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages
(from matplotlib) (4.47.2)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages
(from matplotlib) (1.4.5)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages
(from matplotlib) (23.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packages (from
matplotlib) (9.5.0)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from
datasets) (3.13.1)
Collecting pyarrow>=21.0.0 (from datasets)
  Downloading pyarrow-23.0.1-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (3.1 kB)
Requirement already satisfied: dill<0.4.2,>=0.3.0 in /usr/local/lib/python3.11/dist-packages
(from datasets) (0.3.7)
Collecting requests>=2.32.2 (from datasets)
```

Downloading requests-2.33.1-py3-none-any.whl.metadata (4.8 kB)
Collecting httpx<1.0.0 (from datasets)
 Downloading httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: xxhash in /usr/local/lib/python3.11/dist-packages (from datasets) (3.4.1)
Requirement already satisfied: multiprocessing<0.70.20 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.70.15)
Requirement already satisfied: fsspec<=2026.2.0,>=2023.1.0 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (2023.6.0)
Collecting huggingface-hub<2.0,>=0.25.0 (from datasets)
 Downloading huggingface_hub-1.8.0-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: pyyaml>=5.1 in /usr/lib/python3/dist-packages (from datasets) (5.4.1)
Collecting numpy
 Downloading numpy-1.26.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)

 61.0/61.0 kB 17.6 MB/s eta 0:00:00
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (3.9.1)
Requirement already satisfied: anyio in /usr/local/lib/python3.11/dist-packages (from httpx<1.0.0->datasets) (4.2.0)
Requirement already satisfied: certifi in /usr/lib/python3/dist-packages (from httpx<1.0.0->datasets) (2020.6.20)
Collecting httpcore==1.* (from httpx<1.0.0->datasets)
 Downloading httpcore-1.0.9-py3-none-any.whl.metadata (21 kB)
Requirement already satisfied: idna in /usr/lib/python3/dist-packages (from httpx<1.0.0->datasets) (3.3)
Collecting h11>=0.16 (from httpcore==1.*->httpx<1.0.0->datasets)
 Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting hf-xet<2.0.0,>=1.4.2 (from huggingface-hub<2.0,>=0.25.0->datasets)
 Downloading hf_xet-1.4.3-cp37-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (4.9 kB)
Requirement already satisfied: typer in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (0.9.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (4.9.0)
Requirement already satisfied: six>=1.5 in /usr/lib/python3/dist-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (3.3.2)
Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (2.0.7)
Collecting certifi (from httpx<1.0.0->datasets)
 Downloading certifi-2026.2.25-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (23.1.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (6.0.4)
Requirement already satisfied: yarl<2.0,>=1.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.9.4)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.4.1)
Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.3.1)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.11/dist-packages (from


```

anyio->httpx<1.0.0->datasets) (1.3.0)
Requirement already satisfied: click<9.0.0,>=7.1.1 in /usr/local/lib/python3.11/dist-packages
(from typer->huggingface-hub<2.0,>=0.25.0->datasets) (8.1.7)
Downloading pandas-3.0.2-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (11.3
MB)
_____ 11.3/11.3 MB 147.5 MB/s eta 0:00:0000:010:01
Downloading scikit_learn-1.8.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl
(9.1 MB)
_____ 9.1/9.1 MB 127.5 MB/s eta 0:00:0000:0100:01
Downloading matplotlib-3.10.8-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (8.7
MB)
_____ 8.7/8.7 MB 133.6 MB/s eta 0:00:00a 0:00:01
Downloading tqdm-4.67.3-py3-none-any.whl (78 kB)
_____ 78.4/78.4 kB 24.3 MB/s eta 0:00:00
Downloading datasets-4.8.4-py3-none-any.whl (526 kB)
_____ 527.0/527.0 kB 81.4 MB/s eta 0:00:00
Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)
_____ 84.1/84.1 kB 24.4 MB/s eta 0:00:00
Downloading numpy-1.26.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.3 MB)
_____ 18.3/18.3 MB 72.5 MB/s eta 0:00:00:00:0100:01
Downloading httpx-0.28.1-py3-none-any.whl (73 kB)
_____ 73.5/73.5 kB 21.8 MB/s eta 0:00:00
Downloading httpcore-1.0.9-py3-none-any.whl (78 kB)
_____ 78.8/78.8 kB 22.6 MB/s eta 0:00:00
Downloading huggingface_hub-1.8.0-py3-none-any.whl (625 kB)
_____ 625.2/625.2 kB 89.7 MB/s eta 0:00:00
Downloading pyarrow-23.0.1-cp311-cp311-manylinux_2_28_x86_64.whl (47.6 MB)
_____ 47.6/47.6 MB 64.3 MB/s eta 0:00:00:00:0100:01
Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
_____ 122.8/122.8 kB 33.6 MB/s eta 0:00:00
Downloading requests-2.33.1-py3-none-any.whl (64 kB)
_____ 64.9/64.9 kB 19.6 MB/s eta 0:00:00
Downloading certifi-2026.2.25-py3-none-any.whl (153 kB)
_____ 153.7/153.7 kB 41.4 MB/s eta 0:00:00
Downloading hf_xet-1.4.3-cp37-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (4.2 MB)
_____ 4.2/4.2 MB 144.2 MB/s eta 0:00:00
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Installing collected packages: tqdm, pyparsing, pyarrow, numpy, hf-xet, h11, certifi,
requests, pandas, httpcore, scikit-learn, matplotlib, httpx, huggingface-hub, datasets,
evaluate
  Attempting uninstall: tqdm
    Found existing installation: tqdm 4.66.1
    Uninstalling tqdm-4.66.1:
      Successfully uninstalled tqdm-4.66.1
  Attempting uninstall: pyparsing
    Found existing installation: pyparsing 2.4.7
    Uninstalling pyparsing-2.4.7:
      Successfully uninstalled pyparsing-2.4.7
  Attempting uninstall: pyarrow
    Found existing installation: pyarrow 15.0.0
    Uninstalling pyarrow-15.0.0:
      Successfully uninstalled pyarrow-15.0.0
  Attempting uninstall: numpy
    Found existing installation: numpy 1.26.3
    Uninstalling numpy-1.26.3:

```

```
Successfully uninstalled numpy-1.26.3
Attempting uninstall: certifi
Found existing installation: certifi 2020.6.20
Uninstalling certifi-2020.6.20:
Successfully uninstalled certifi-2020.6.20
Attempting uninstall: requests
Found existing installation: requests 2.31.0
Uninstalling requests-2.31.0:
Successfully uninstalled requests-2.31.0
Attempting uninstall: pandas
Found existing installation: pandas 2.2.0
Uninstalling pandas-2.2.0:
Successfully uninstalled pandas-2.2.0
Attempting uninstall: scikit-learn
Found existing installation: scikit-learn 1.3.0
Uninstalling scikit-learn-1.3.0:
Successfully uninstalled scikit-learn-1.3.0
Attempting uninstall: matplotlib
Found existing installation: matplotlib 3.7.3
Uninstalling matplotlib-3.7.3:
Successfully uninstalled matplotlib-3.7.3
Attempting uninstall: huggingface-hub
Found existing installation: huggingface-hub 0.20.3
Uninstalling huggingface-hub-0.20.3:
Successfully uninstalled huggingface-hub-0.20.3
Attempting uninstall: datasets
Found existing installation: datasets 2.14.5
Uninstalling datasets-2.14.5:
Successfully uninstalled datasets-2.14.5
ERROR: pip's dependency resolver does not currently take into account all the packages that
are installed. This behaviour is the source of the following dependency conflicts.
gradient 2.0.6 requires attrs<=19, but you have attrs 23.1.0 which is incompatible.
transformers 4.35.2 requires huggingface-hub<1.0,>=0.16.4, but you have huggingface-hub 1.8.0
which is incompatible.
tokenizers 0.15.1 requires huggingface_hub<1.0,>=0.16.4, but you have huggingface-hub 1.8.0
which is incompatible.
Successfully installed certifi-2026.2.25 datasets-4.8.4 evaluate-0.4.6 h11-0.16.0 hf-
xet-1.4.3 httpcore-1.0.9 httpx-0.28.1 huggingface-hub-1.8.0 matplotlib-3.10.8 numpy-1.26.4
pandas-3.0.2 pyarrow-23.0.1 pyparsing-3.3.2 requests-2.33.1 scikit-learn-1.8.0 tqdm-4.67.3
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting
behaviour with the system package manager. It is recommended to use a virtual environment
instead: https://pip.pypa.io/warnings/venv
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: transformers in /usr/local/lib/python3.11/dist-packages
(4.35.2)
Collecting transformers
  Downloading transformers-5.4.0-py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: accelerate in /usr/local/lib/python3.11/dist-packages (0.24.1)
Collecting accelerate
  Downloading accelerate-1.13.0-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.11/dist-packages
(0.1.99)
Collecting sentencepiece
  Downloading sentencepiece-0.2.1-cp311-cp311-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (10 kB)
```

Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.11/dist-packages (from transformers) (1.8.0)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from transformers) (1.26.4)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from transformers) (23.2)

Requirement already satisfied: pyyaml>=5.1 in /usr/lib/python3/dist-packages (from transformers) (5.4.1)

Collecting regex>=2025.10.22 (from transformers)

 Downloading regex-2026.3.32-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (40 kB)

40.4/40.4 kB 10.5 MB/s eta 0:00:00

Collecting tokenizers<=0.23.0,>=0.22.0 (from transformers)

 Downloading tokenizers-0.22.2-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.3 kB)

Requirement already satisfied: typer in /usr/local/lib/python3.11/dist-packages (from transformers) (0.9.0)

Collecting safetensors>=0.4.3 (from transformers)

 Downloading safetensors-0.7.0-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)

Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.11/dist-packages (from transformers) (4.67.3)

Requirement already satisfied: psutil in /usr/local/lib/python3.11/dist-packages (from accelerate) (5.9.8)

Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from accelerate) (2.1.1+cu121)

Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (3.13.1)

Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (2023.6.0)

Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (1.4.3)

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0.28.1)

Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (4.9.0)

Requirement already satisfied: sympy in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (1.12)

Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (3.2.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (3.1.3)

Requirement already satisfied: triton==2.1.0 in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (2.1.0)

Requirement already satisfied: click<9.0.0,>=7.1.1 in /usr/local/lib/python3.11/dist-packages (from typer->transformers) (8.1.7)

Requirement already satisfied: anyio in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (4.2.0)

Requirement already satisfied: certifi in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (2026.2.25)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.0.9)

Requirement already satisfied: idna in /usr/lib/python3/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (3.3)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.11/dist-packages (from

httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages
(from Jinja2->torch>=2.0.0->accelerate) (2.1.4)
Requirement already satisfied: mpmath>=0.19 in /usr/local/lib/python3.11/dist-packages (from
sympy->torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.11/dist-packages (from
anyio->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.3.0)
Downloading transformers-5.4.0-py3-none-any.whl (10.1 MB)

10.1/10.1 MB 114.8 MB/s eta 0:00:0000:01:01

Downloading accelerate-1.13.0-py3-none-any.whl (383 kB)

383.7/383.7 kB 77.2 MB/s eta 0:00:00

Downloading sentencepiece-0.2.1-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl
(1.4 MB)

1.4/1.4 MB 112.9 MB/s eta 0:00:00

Downloading regex-2026.3.32-cp311-cp311-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (798 kB)

798.3/798.3 kB 94.8 MB/s eta 0:00:00

Downloading safetensors-0.7.0-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (507
kB)

507.2/507.2 kB 79.8 MB/s eta 0:00:00

Downloading tokenizers-0.22.2-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.3
MB)

3.3/3.3 MB 138.3 MB/s eta 0:00:00

Installing collected packages: sentencepiece, safetensors, regex, tokenizers, accelerate,
transformers

Attempting uninstall: sentencepiece

Found existing installation: sentencepiece 0.1.99

Uninstalling sentencepiece-0.1.99:

Successfully uninstalled sentencepiece-0.1.99

Attempting uninstall: safetensors

Found existing installation: safetensors 0.4.0

Uninstalling safetensors-0.4.0:

Successfully uninstalled safetensors-0.4.0

Attempting uninstall: regex

Found existing installation: regex 2023.12.25

Uninstalling regex-2023.12.25:

Successfully uninstalled regex-2023.12.25

Attempting uninstall: tokenizers

Found existing installation: tokenizers 0.15.1

Uninstalling tokenizers-0.15.1:

Successfully uninstalled tokenizers-0.15.1

Attempting uninstall: accelerate

Found existing installation: accelerate 0.24.1

Uninstalling accelerate-0.24.1:

Successfully uninstalled accelerate-0.24.1

Attempting uninstall: transformers

Found existing installation: transformers 4.35.2

Uninstalling transformers-4.35.2:

Successfully uninstalled transformers-4.35.2

ERROR: pip's dependency resolver does not currently take into account all the packages that
are installed. This behaviour is the source of the following dependency conflicts.
sentence-transformers 2.2.2 requires transformers<5.0.0,>=4.6.0, but you have transformers
5.4.0 which is incompatible.

Successfully installed accelerate-1.13.0 regex-2026.3.32 safetensors-0.7.0
sentencepiece-0.2.1 tokenizers-0.22.2 transformers-5.4.0

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>

Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: bitsandbytes in /usr/local/lib/python3.11/dist-packages (0.41.2)

Collecting bitsandbytes

Downloading bitsandbytes-0.49.2-py3-none-manylinux_2_24_x86_64.whl.metadata (10 kB)

Collecting torch<3,>=2.3 (from bitsandbytes)

Downloading torch-2.11.0-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (29 kB)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from bitsandbytes) (1.26.4)

Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.11/dist-packages (from bitsandbytes) (23.2)

Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (3.13.1)

Collecting typing-extensions>=4.10.0 (from torch<3,>=2.3->bitsandbytes)

Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (69.0.3)

Collecting sympy>=1.13.3 (from torch<3,>=2.3->bitsandbytes)

Downloading sympy-1.14.0-py3-none-any.whl.metadata (12 kB)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (3.2.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (3.1.3)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (2023.6.0)

Collecting cuda-toolkit==13.0.2 (from cuda-

toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)

Downloading cuda_toolkit-13.0.2-py2.py3-none-any.whl.metadata (9.4 kB)

Collecting cuda-bindings<14,>=13.0.3 (from torch<3,>=2.3->bitsandbytes)

Downloading cuda_bindings-13.2.0-cp311-cp311-

manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (2.3 kB)

Collecting nvidia-cudnn-cu13==9.19.0.56 (from torch<3,>=2.3->bitsandbytes)

Downloading nvidia_cudnn_cu13-9.19.0.56-py3-none-manylinux_2_27_x86_64.whl.metadata (1.9 kB)

Collecting nvidia-cusparselt-cu13==0.8.0 (from torch<3,>=2.3->bitsandbytes)

Downloading nvidia_cusparselt_cu13-0.8.0-py3-none-manylinux2014_x86_64.whl.metadata (12 kB)

Collecting nvidia-nccl-cu13==2.28.9 (from torch<3,>=2.3->bitsandbytes)

Downloading nvidia_nccl_cu13-2.28.9-py3-none-manylinux_2_18_x86_64.whl.metadata (2.0 kB)

Collecting nvidia-nvshmem-cu13==3.4.5 (from torch<3,>=2.3->bitsandbytes)

Downloading nvidia_nvshmem_cu13-3.4.5-py3-none-

manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.1 kB)

Collecting triton==3.6.0 (from torch<3,>=2.3->bitsandbytes)

Downloading triton-3.6.0-cp311-cp311-

manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (1.7 kB)

Collecting nvidia-cublas==13.1.0.3.* (from cuda-

toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)

Downloading nvidia_cublas-13.1.0.3-py3-none-manylinux_2_27_x86_64.whl.metadata (1.7 kB)

Collecting nvidia-cuda-runtime==13.0.96.* (from cuda-

toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)

Downloading nvidia_cuda_runtime-13.0.96-py3-none-manylinux2014_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cufft==12.0.0.61.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
Downloading nvidia_cufft-12.0.0.61-py3-none-manylinux2014_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cufile==1.15.1.6.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
Downloading nvidia_cufile-1.15.1.6-py3-none-manylinux2014_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-cupti==13.0.85.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
Downloading nvidia_cuda_cupti-13.0.85-py3-none-manylinux_2_25_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-curand==10.4.0.35.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
Downloading nvidia_curand-10.4.0.35-py3-none-manylinux_2_27_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cusolver==12.0.4.66.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
Downloading nvidia_cusolver-12.0.4.66-py3-none-manylinux_2_27_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cusparsenvjitlink==13.0.88.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
Downloading nvidia_cusparsenvjitlink-13.0.88-py3-none-manylinux2014_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-nvjitlink==13.0.88.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
Downloading nvidia_nvjitlink-13.0.88-py3-none-manylinux2010_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-nvrtc==13.0.88.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-nvtx==13.0.85.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
Downloading nvidia_nvtx-13.0.85-py3-none-manylinux1_x86_64.whl.metadata (1.8 kB)
Collecting cuda-pathfinder~1.1 (from cuda-bindings<14,>=13.0.3->torch<3,>=2.3->bitsandbytes)
Downloading cuda_pathfinder-1.5.0-py3-none-any.whl.metadata (1.9 kB)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy>=1.13.3->torch<3,>=2.3->bitsandbytes) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from jinja2->torch<3,>=2.3->bitsandbytes) (2.1.4)
Downloading bitsandbytes-0.49.2-py3-none-manylinux_2_24_x86_64.whl (60.7 MB)
60.7/60.7 MB 50.2 MB/s eta 0:00:00:00:0100:01
Downloading torch-2.11.0-cp311-cp311-manylinux_2_28_x86_64.whl (530.6 MB)
530.6/530.6 MB 7.9 MB/s eta 0:00:00:00:0100:01
Downloading cuda_toolkit-13.0.2-py2.py3-none-any.whl (2.4 kB)

```
Downloading nvidia_cudnn_cu13-9.19.0.56-py3-none-manylinux_2_27_x86_64.whl (366.1 MB)
_____ 366.1/366.1 MB 10.1 MB/s eta 0:00:000:0100:01
Downloading nvidia_cusparselt_cu13-0.8.0-py3-none-manylinux2014_x86_64.whl (169.9 MB)
_____ 169.9/169.9 MB 22.6 MB/s eta 0:00:000:0100:01
Downloading nvidia_nccl_cu13-2.28.9-py3-none-manylinux_2_18_x86_64.whl (196.5 MB)
_____ 196.5/196.5 MB 21.0 MB/s eta 0:00:000:0100:01
Downloading nvidia_nvshmem_cu13-3.4.5-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (60.4 MB)
_____ 60.4/60.4 MB 54.2 MB/s eta 0:00:00:00:0100:01
Downloading triton-3.6.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (188.2 MB)
_____ 188.2/188.2 MB 20.3 MB/s eta 0:00:000:0100:01
Downloading nvidia_cublas-13.1.0.3-py3-none-manylinux_2_27_x86_64.whl (423.1 MB)
_____ 423.1/423.1 MB 9.0 MB/s eta 0:00:00:00:0100:01
Downloading nvidia_cuda_cupti-13.0.85-py3-none-manylinux_2_25_x86_64.whl (10.7 MB)
_____ 10.7/10.7 MB 94.6 MB/s eta 0:00:00:00:010:01
Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl (90.2 MB)
_____ 90.2/90.2 MB 35.7 MB/s eta 0:00:00:00:0100:01
Downloading nvidia_cuda_runtime-13.0.96-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (2.2 MB)
_____ 2.2/2.2 MB 49.9 MB/s eta 0:00:00a 0:00:01
Downloading nvidia_cufft-12.0.0.61-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (214.1 MB)
_____ 214.1/214.1 MB 12.0 MB/s eta 0:00:000:0100:01
Downloading nvidia_cufile-1.15.1.6-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1.2 MB)
_____ 1.2/1.2 MB 74.8 MB/s eta 0:00:00
Downloading nvidia_curand-10.4.0.35-py3-none-manylinux_2_27_x86_64.whl (59.5 MB)
_____ 59.5/59.5 MB 55.1 MB/s eta 0:00:00:00:0100:01
Downloading nvidia_cusolver-12.0.4.66-py3-none-manylinux_2_27_x86_64.whl (200.9 MB)
_____ 200.9/200.9 MB 17.9 MB/s eta 0:00:000:0100:01
Downloading nvidia_cusparselt-12.6.3.3-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (145.9 MB)
_____ 145.9/145.9 MB 27.5 MB/s eta 0:00:000:0100:01
Downloading nvidia_nvjitlink-13.0.88-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl (40.7 MB)
_____ 40.7/40.7 MB 80.0 MB/s eta 0:00:00:00:0100:01
Downloading nvidia_nvtx-13.0.85-py3-none-manylinux1_x86_64.manylinux_2_5_x86_64.whl (148 kB)
_____ 148.0/148.0 kB 41.6 MB/s eta 0:00:00
Downloading cuda_bindings-13.2.0-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (6.3 MB)
_____ 6.3/6.3 MB 162.0 MB/s eta 0:00:000:00:01
Downloading sympy-1.14.0-py3-none-any.whl (6.3 MB)
_____ 6.3/6.3 MB 101.2 MB/s eta 0:00:000:00:01
Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
_____ 44.6/44.6 kB 3.2 MB/s eta 0:00:00
Downloading cuda_pathfinder-1.5.0-py3-none-any.whl (49 kB)
_____ 49.7/49.7 kB 4.9 MB/s eta 0:00:00
```

Installing collected packages: nvidia-cusparselt-cu13, cuda-toolkit, typing-extensions, triton, sympy, nvidia-nvtx, nvidia-nvshmem-cu13, nvidia-nvjitlink, nvidia-nccl-cu13, nvidia-curand, nvidia-cufile, nvidia-cuda-runtime, nvidia-cuda-nvrtc, nvidia-cuda-cupti, nvidia-cublas, cuda-pathfinder, nvidia-cusparselt, nvidia-cufft, nvidia-cudnn-cu13, cuda-bindings, nvidia-cusolver, torch, bitsandbytes

Attempting uninstall: typing-extensions

```

Found existing installation: typing_extensions 4.9.0
Uninstalling typing_extensions-4.9.0:
  Successfully uninstalled typing_extensions-4.9.0
Attempting uninstall: triton
  Found existing installation: triton 2.1.0
  Uninstalling triton-2.1.0:
    Successfully uninstalled triton-2.1.0
Attempting uninstall: sympy
  Found existing installation: sympy 1.12
  Uninstalling sympy-1.12:
    Successfully uninstalled sympy-1.12
Attempting uninstall: torch
  Found existing installation: torch 2.1.1+cu121
  Uninstalling torch-2.1.1+cu121:
    Successfully uninstalled torch-2.1.1+cu121
Attempting uninstall: bitsandbytes
  Found existing installation: bitsandbytes 0.41.2
  Uninstalling bitsandbytes-0.41.2:
    Successfully uninstalled bitsandbytes-0.41.2
ERROR: pip's dependency resolver does not currently take into account all the packages that
are installed. This behaviour is the source of the following dependency conflicts.
sentence-transformers 2.2.2 requires transformers<5.0.0,>=4.6.0, but you have transformers
5.4.0 which is incompatible.
torchaudio 2.1.1+cu121 requires torch==2.1.1, but you have torch 2.11.0 which is
incompatible.
torchvision 0.16.1+cu121 requires torch==2.1.1, but you have torch 2.11.0 which is
incompatible.
Successfully installed bitsandbytes-0.49.2 cuda-bindings-13.2.0 cuda-pathfinder-1.5.0 cuda-
toolkit-13.0.2 nvidia-cublas-13.1.0.3 nvidia-cuda-cupti-13.0.85 nvidia-cuda-nvrtc-13.0.88
nvidia-cuda-runtime-13.0.96 nvidia-cudnn-cu13-9.19.0.56 nvidia-cufft-12.0.0.61 nvidia-
cufile-1.15.1.6 nvidia-curand-10.4.0.35 nvidia-cusolver-12.0.4.66 nvidia-cusparselt-12.6.3.3
nvidia-cusparse-12.6.3.3 nvidia-cusparselt-cu13-0.8.0 nvidia-nccl-cu13-2.28.9 nvidia-nvjitlink-13.0.88 nvidia-nvshmem-
cu13-3.4.5 nvidia-nvtx-13.0.85 sympy-1.14.0 torch-2.11.0 triton-3.6.0 typing-
extensions-4.15.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting
behaviour with the system package manager. It is recommended to use a virtual environment
instead: https://pip.pypa.io/warnings/venv
Note: you may need to restart the kernel to use updated packages.
Found existing installation: torch 2.11.0
Uninstalling torch-2.11.0:
  Successfully uninstalled torch-2.11.0
Found existing installation: torchvision 0.16.1+cu121
Uninstalling torchvision-0.16.1+cu121:
  Successfully uninstalled torchvision-0.16.1+cu121
Found existing installation: torchaudio 2.1.1+cu121
Uninstalling torchaudio-2.1.1+cu121:
  Successfully uninstalled torchaudio-2.1.1+cu121
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting
behaviour with the system package manager. It is recommended to use a virtual environment
instead: https://pip.pypa.io/warnings/venv
Note: you may need to restart the kernel to use updated packages.
ERROR: Invalid requirement: 'nvidia-*'
Note: you may need to restart the kernel to use updated packages.
WARNING: Skipping nvidia-cuda-runtime-cu12 as it is not installed.
WARNING: Skipping nvidia-cublas-cu12 as it is not installed.

```


WARNING: Skipping nvidia-cusparse-cu12 as it is not installed.
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>

Note: you may need to restart the kernel to use updated packages.

Looking in indexes: <https://download.pytorch.org/whl/cu118>

Collecting torch

Downloading https://download.pytorch.org/whl/cu118/torch-2.7.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (28 kB)

Collecting torchvision

Downloading https://download-r2.pytorch.org/whl/cu118/torchvision-0.22.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (6.1 kB)

Collecting torchaudio

Downloading https://download-r2.pytorch.org/whl/cu118/torchaudio-2.7.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (6.6 kB)

Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from torch) (3.13.1)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.11/dist-packages (from torch) (4.15.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.11/dist-packages (from torch) (1.14.0)

Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch) (3.2.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch) (3.1.3)

Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (from torch) (2023.6.0)

Collecting nvidia-cuda-nvrtc-cu11==11.8.89 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cuda_nvrtc_cu11-11.8.89-py3-none-manylinux1_x86_64.whl (23.2 MB)

23.2/23.2 MB 106.5 MB/s eta 0:00:000:0100:01

Collecting nvidia-cuda-runtime-cu11==11.8.89 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cuda_runtime_cu11-11.8.89-py3-none-manylinux1_x86_64.whl (875 kB)

875.6/875.6 kB 105.0 MB/s eta 0:00:00

Collecting nvidia-cuda-cupti-cu11==11.8.87 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cuda_cupti_cu11-11.8.87-py3-none-manylinux1_x86_64.whl (13.1 MB)

13.1/13.1 MB 152.7 MB/s eta 0:00:000:0100:01

Collecting nvidia-cudnn-cu11==9.1.0.70 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cudnn_cu11-9.1.0.70-py3-none-manylinux2014_x86_64.whl (663.9 MB)

663.9/663.9 MB 4.9 MB/s eta 0:00:000:0100:01

Collecting nvidia-cublas-cu11==11.11.3.6 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cublas_cu11-11.11.3.6-py3-none-manylinux1_x86_64.whl (417.9 MB)

417.9/417.9 MB 8.9 MB/s eta 0:00:000:0100:01

Collecting nvidia-cufft-cu11==10.9.0.58 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cufft_cu11-10.9.0.58-py3-none-manylinux1_x86_64.whl (168.4 MB)

168.4/168.4 MB 23.0 MB/s eta 0:00:000:0100:01

Collecting nvidia-curand-cu11==10.3.0.86 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_curand_cu11-10.3.0.86-py3-none-manylinux1_x86_64.whl (58.1 MB)

58.1/58.1 MB 55.4 MB/s eta 0:00:000:0100:01

```

Collecting nvidia-cusolver-cu11==11.4.1.48 (from torch)
  Downloading https://download.pytorch.org/whl/cu118/nvidia_cusolver_cu11-11.4.1.48-py3-none-manylinux1_x86_64.whl (128.2 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 128.2/128.2 MB 23.3 MB/s eta 0:00:000:0100:01
Collecting nvidia-cuspars-cu11==11.7.5.86 (from torch)
  Downloading https://download.pytorch.org/whl/cu118/nvidia_cuspars-cu11-11.7.5.86-py3-none-manylinux1_x86_64.whl (204.1 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 204.1/204.1 MB 16.9 MB/s eta 0:00:000:0100:01
Collecting nvidia-nccl-cu11==2.21.5 (from torch)
  Downloading https://download.pytorch.org/whl/cu118/nvidia_nccl_cu11-2.21.5-py3-none-manylinux2014_x86_64.whl (147.8 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 147.8/147.8 MB 23.1 MB/s eta 0:00:000:0100:01
Collecting nvidia-nvtx-cu11==11.8.86 (from torch)
  Downloading https://download.pytorch.org/whl/cu118/nvidia_nvtx_cu11-11.8.86-py3-none-manylinux1_x86_64.whl (99 kB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 99.1/99.1 kB 23.9 MB/s eta 0:00:00
Collecting triton==3.3.1 (from torch)
  Downloading https://download-r2.pytorch.org/whl/triton-3.3.1-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (1.5 kB)
Requirement already satisfied: setuptools>=40.8.0 in /usr/local/lib/python3.11/dist-packages (from triton==3.3.1->torch) (69.0.3)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from torchvision) (1.26.4)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /usr/local/lib/python3.11/dist-packages (from torchvision) (9.5.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from jinja2->torch) (2.1.4)
Downloading https://download.pytorch.org/whl/cu118/torch-2.7.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl (905.3 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 905.3/905.3 MB 4.8 MB/s eta 0:00:00:00:0100:01
Downloading https://download-r2.pytorch.org/whl/triton-3.3.1-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (155.7 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 155.7/155.7 MB 24.6 MB/s eta 0:00:000:0100:01
Downloading https://download-r2.pytorch.org/whl/cu118/torchvision-0.22.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl (6.7 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.7/6.7 MB 100.1 MB/s eta 0:00:000:00:01
Downloading https://download-r2.pytorch.org/whl/cu118/torchaudio-2.7.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl (3.3 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.3/3.3 MB 67.7 MB/s eta 0:00:00a 0:00:01
Installing collected packages: triton, nvidia-nvtx-cu11, nvidia-nccl-cu11, nvidia-cuspars-cu11, nvidia-curand-cu11, nvidia-cufft-cu11, nvidia-cuda-runtime-cu11, nvidia-cuda-nvrtc-cu11, nvidia-cuda-cupti-cu11, nvidia-cublas-cu11, nvidia-cusolver-cu11, nvidia-cudnn-cu11, torch, torchvision, torchaudio
Attempting uninstall: triton
  Found existing installation: triton 3.6.0
  Uninstalling triton-3.6.0:
    Successfully uninstalled triton-3.6.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
sentence-transformers 2.2.2 requires transformers<5.0.0,>=4.6.0, but you have transformers 5.4.0 which is incompatible.
Successfully installed nvidia-cublas-cu11-11.11.3.6 nvidia-cuda-cupti-cu11-11.8.87 nvidia-cuda-nvrtc-cu11-11.8.89 nvidia-cuda-runtime-cu11-11.8.89 nvidia-cudnn-cu11-9.1.0.70 nvidia-

```

cufft-cu11-10.9.0.58 nvidia-cusolver-cu11-11.4.1.48 nvidia-cusparse-cu11-11.7.5.86 nvidia-nccl-cu11-2.21.5 nvidia-nvtx-cu11-11.8.86 torch-2.7.1+cu118 torchaudio-2.7.1+cu118 torchvision-0.22.1+cu118 triton-3.3.1

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>

Note: you may need to restart the kernel to use updated packages.

```
In [7]: # JupyterLab / fresh GPU machine setup
import sys
print(sys.executable)
```

```
Out[7]: /usr/local/bin/python3
```

Import Required Libraries

Set Random Seed for Reproducibility

```
In [1]: import os
import re
import random
import numpy as np
import pandas as pd
import torch

from tqdm import tqdm
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix, classification_report
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(SEED)
```

Upload Dataset

Upload **sample_full.csv** from the Monash dataset.

The dataset contains:

- Learning_outcome
- Bloom taxonomy one-hot columns

```
In [2]: import os
import pandas as pd

CSV_PATH = "content/sample_full.csv" # apni CSV ka exact naam do
assert os.path.exists(CSV_PATH), f"CSV file not found: {CSV_PATH}"

df = pd.read_csv(CSV_PATH)
print("Dataset size:", len(df))
df.head()
```

Out[2]: Dataset size: 21380

Out[2]:

	Learning_outcome	Remember	Understand	Apply	Analyze	Evaluate	Create
0	Analyze the health economic implications of e...	NaN	NaN	NaN	1.0	NaN	NaN
1	Apply research skills to operate effectively ...	NaN	NaN	1.0	NaN	NaN	NaN
2	Assess and synthesise diverse information abo...	NaN	NaN	NaN	NaN	1.0	1.0
3	Describe the general characteristics of the m...	NaN	1.0	NaN	NaN	NaN	NaN
4	Evaluate the different models of perioperativ...	NaN	NaN	NaN	NaN	1.0	NaN

Convert One-Hot Encoding to Single Label

Bloom taxonomy columns:

- Remember
- Understand
- Apply
- Analyze
- Evaluate
- Create

```
In [3]: LABELS = ["Remember", "Understand", "Apply", "Analyze", "Evaluate", "Create"]

required_cols = ["Learning_outcome"] + LABELS
missing_cols = [c for c in required_cols if c not in df.columns]
assert not missing_cols, f"Missing columns: {missing_cols}"

df["label"] = df[LABELS].idxmax(axis=1)
df = df[["Learning_outcome", "label"]].dropna().reset_index(drop=True)

print(df.head())
print(df["label"].value_counts())
```

	Learning_outcome	label
0	Analyze the health economic implications of e...	Analyze
1	Apply research skills to operate effectively ...	Apply
2	Assess and synthesise diverse information abo...	Evaluate
3	Describe the general characteristics of the m...	Understand
4	Evaluate the different models of perioperativ...	Evaluate

label

Apply	5801
Understand	5694
Analyze	3096
Create	2955
Evaluate	2649
Remember	1185

Name: count, dtype: int64

Train / Test Split

80% training data

20% testing data

```
In [4]: train_df, test_df = train_test_split(
        df,
        test_size=0.2,
        random_state=SEED,
        stratify=df["label"]
    )

print("Train size:", len(train_df))
print("Test size:", len(test_df))
```

```
Out[4]: Train size: 17104
        Test size: 4276
```

Load Model (4-bit Quantization)

4-bit quantization allows large models to run on Colab T4 GPU.

```
In [5]: import torch
import platform

print("Python:", platform.python_version())
print("PyTorch:", torch.__version__)
print("CUDA available:", torch.cuda.is_available())

if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))
    print("CUDA device count:", torch.cuda.device_count())
    print("Current device:", torch.cuda.current_device())
```

```
else:
    print("Running on CPU")
```

Out[5]:

```
Python: 3.11.7
PyTorch: 2.7.1+cu118
CUDA available: True
GPU: NVIDIA RTX A6000
CUDA device count: 1
Current device: 0
```

In [6]:

```
import torch
if torch.cuda.is_available():
    torch.cuda.empty_cache()
    print("CUDA cache cleared.")
else:
    print("CUDA is not available, no cache to clear.")
```

Out[6]:

```
CUDA cache cleared.
```

In [7]:

```
from huggingface_hub import login

HF_TOKEN = "hf_byvpkLIEiQdmtaXnvSRdYseb1lNHbXtnPs"

login(HF_TOKEN)
```

Model Registry

In [8]:

```
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig

MODEL_REGISTRY = {
    # Already using
    "mistral_7b_instruct_v02": "mistralai/Mistral-7B-Instruct-v0.3",
    "llama3_8b_instruct": "meta-llama/Llama-3.1-8B-Instruct",
    "phi35_mini_4k_instruct": "microsoft/Phi-3.5-mini-instruct",
    "qwen2_7b_instruct": "Qwen/Qwen2-7B-Instruct",

    # Recommended additions
    "llama31_8b_instruct": "meta-llama/Llama-3.1-8B-Instruct",
    "qwen25_7b_instruct": "Qwen/Qwen2.5-7B-Instruct",

    # Smaller / faster options
    "qwen25_3b_instruct": "Qwen/Qwen2.5-3B-Instruct",
    "qwen25_1_5b_instruct": "Qwen/Qwen2.5-1.5B-Instruct",
    "smollm2_1_7b_instruct": "HuggingFaceTB/SmolLM2-1.7B-Instruct",
    "gemma2_2b_instruct": "google/gemma-2-2b-it",
    "gemma2_9b_instruct": "google/gemma-2-9b-it",

    # Good experimental middle tier
    "phi35_mini_instruct": "microsoft/Phi-4-mini-instruct",
```

```
"mistral_nemo_instruct": "mistralai/Mistral-Nemo-Instruct-2407"
}
```

In [9]:

```
def load_model_and_tokenizer(model_id):

    bnb_config = BitsAndBytesConfig(
        load_in_4bit=True,
        bnb_4bit_quant_type="nf4",
        bnb_4bit_use_double_quant=True,
        bnb_4bit_compute_dtype=torch.float16
    )

    tokenizer = AutoTokenizer.from_pretrained(model_id, use_fast=False)

    # PAD token fix
    if tokenizer.pad_token is None:
        tokenizer.pad_token = tokenizer.eos_token

    model = AutoModelForCausalLM.from_pretrained(
        model_id,
        quantization_config=bnb_config,
        device_map="auto",
        torch_dtype=torch.float16
    )

    model.config.pad_token_id = tokenizer.pad_token_id

    return tokenizer, model
```

Prompt Design

We use a **zero-shot classification prompt**.

The model must return only one Bloom taxonomy label.

In [10]:

```
LABELS = ["Remember", "Understand", "Apply", "Analyze", "Evaluate", "Create"]
```

```
def create_prompt(text):
    return f"""You are a strict Bloom's Taxonomy classifier.
```

```
Classify the given learning outcome into exactly one of these labels:
Remember, Understand, Apply, Analyze, Evaluate, Create
```

```
Return only the label name.
Do not explain.
Do not write a sentence.
```

```
Examples:
```

```
Learning Outcome: Define the basic concepts of machine learning.
Label: Remember
```


Learning Outcome: Explain the difference between supervised and unsupervised learning.

Label: Understand

Learning Outcome: Use Python to implement a linear regression model.

Label: Apply

Learning Outcome: Differentiate between overfitting and underfitting in a given model.

Label: Analyze

Learning Outcome: Critique the performance of a classification model using evaluation metrics.

Label: Evaluate

Learning Outcome: Design an intelligent system for detecting fake news.

Label: Create

Now classify this:

Learning Outcome: {text}

Label: ""

Normalize Model Output

In [11]:

```
LABELS = ["Remember", "Understand", "Apply", "Analyze", "Evaluate", "Create"]

def normalize_output(text):
    text = str(text).strip().lower()

    matches = []

    for label in LABELS:
        pattern = r"\b" + re.escape(label.lower()) + r"\b"
        if re.search(pattern, text):
            matches.append(label)

    if matches:
        return matches[-1]

    return "Invalid"
```

LLM Prediction Function

In [12]:

```
def predict(text, tokenizer, model, max_new_tokens=12):
    prompt = create_prompt(text)

    inputs = tokenizer(
        prompt,
        return_tensors="pt",
        truncation=True,
        max_length=512
```

```

).to(model.device)

input_length = inputs["input_ids"].shape[1]

with torch.no_grad():
    output = model.generate(
        **inputs,
        max_new_tokens=max_new_tokens,
        do_sample=False,
        pad_token_id=tokenizer.pad_token_id
    )

generated_tokens = output[0][input_length:]
decoded = tokenizer.decode(generated_tokens, skip_special_tokens=True).strip()

return decoded

```

Evaluate Model on Test Data

In [13]:

```

import gc
import torch

def clear_memory():
    gc.collect()
    if torch.cuda.is_available():
        torch.cuda.empty_cache()

```

In [14]:

```

sample_model_name = list(MODEL_REGISTRY.keys())[0]
sample_model_id = MODEL_REGISTRY[sample_model_name]

print("Testing model:", sample_model_name)

tokenizer, model = load_model_and_tokenizer(sample_model_id)

for i in range(10):
    text = test_df.iloc[i]["Learning_outcome"]
    raw_pred = predict(text, tokenizer, model)
    clean_pred = normalize_output(raw_pred)

    print("=" * 80)
    print("TEXT:", text)
    print("RAW OUTPUT:", raw_pred)
    print("NORMALIZED:", clean_pred)

del model
del tokenizer
clear_memory()

```

Out[14]: Testing model: mistral_7b_instruct_v02

Out[14]:

config.json: 0%| | 0.00/601 [00:00<?, ?B/s]

Out[14]:

tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[14]:

tokenizer.json: 0.00B [00:00, ?B/s]

Out[14]:

tokenizer.model: 0%| | 0.00/587k [00:00<?, ?B/s]

Out[14]:

special_tokens_map.json: 0%| | 0.00/414 [00:00<?, ?B/s]

Out[14]:

`torch\_dtype` is deprecated! Use `dtype` instead!

Out[14]:

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[14]:

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[14]:

Fetching 3 files: 0%| | 0/3 [00:00<?, ?it/s]

Out[14]:

Loading weights: 0%| | 0/291 [00:00<?, ?it/s]

Out[14]:

generation_config.json: 0%| | 0.00/116 [00:00<?, ?B/s]

Out[14]:

```
=====
TEXT: appreciate science and technologies as a way of thinking and acting that generates
knowledge and which represents cultures and societies
RAW OUTPUT: Understand
NORMALIZED: Understand
=====
TEXT: Critically reflect on your own design practice and projects to date;
RAW OUTPUT: Evaluate
NORMALIZED: Evaluate
=====
TEXT: Reflect on the impact of recent developments in science and philosophy on our most
fundamental views about ourselves and our place in the world;
RAW OUTPUT: Evaluate
NORMALIZED: Evaluate
=====
TEXT: Integrate and summarise information obtained during psychological assessments in the
form of a professional report;
RAW OUTPUT: Apply
NORMALIZED: Apply
=====
TEXT: Have developed the ability to apply philosophical theories of justice in conflict to
actual cases from recent history;
RAW OUTPUT: Apply
NORMALIZED: Apply
```

```

=====
TEXT: Describe medical and professional ethics and different ethical perspectives as related
to healthcare
RAW OUTPUT: Remember
NORMALIZED: Remember
=====
TEXT: Conduct research in topics covered in Current Issues in Workplace Law based on
knowledge of appropriate research principles and methods
RAW OUTPUT: Apply
NORMALIZED: Apply
=====
TEXT: Describe and implement patient care and quality assurance principles to sonographic
imaging.
RAW OUTPUT: Apply
NORMALIZED: Apply
=====
TEXT: Demonstrate the ability to research and write from an informed position about the body
in contemporary art.
RAW OUTPUT: Understand
NORMALIZED: Understand
=====
TEXT: Identify data resources and standards.
RAW OUTPUT: Remember
NORMALIZED: Remember

```

In [15]:

```

from sklearn.metrics import (
    accuracy_score,
    f1_score,
    precision_score,
    recall_score,
    confusion_matrix,
    classification_report,
    balanced_accuracy_score,
    matthews_corrcoef,
    cohen_kappa_score
)

all_results = []
all_predictions = {}

for model_name, model_id in MODEL_REGISTRY.items():
    print("=" * 80)
    print(f"Loading model: {model_name} -> {model_id}")

    if torch.cuda.is_available():
        torch.cuda.empty_cache()

    tokenizer, model = None, None

    try:
        tokenizer, model = load_model_and_tokenizer(model_id)

        predictions = []
        for _, row in tqdm(test_df.iterrows(), total=len(test_df), desc=model_name):

```

```

        pred = predict(row["Learning_outcome"], tokenizer, model)
        predictions.append(pred)

result_df = test_df.copy()
result_df["raw_prediction"] = predictions
result_df["prediction"] = result_df["raw_prediction"].apply(normalize_output)

result_df = result_df.dropna(subset=["label", "prediction"])
y_true = result_df["label"].astype(str).str.strip()
y_pred = result_df["prediction"].astype(str).str.strip()

accuracy = accuracy_score(y_true, y_pred)
macro_f1 = f1_score(y_true, y_pred, average="macro", zero_division=0)
micro_f1 = f1_score(y_true, y_pred, average="micro", zero_division=0)
weighted_f1 = f1_score(y_true, y_pred, average="weighted", zero_division=0)
macro_precision = precision_score(y_true, y_pred, average="macro", zero_division=0)
macro_recall = recall_score(y_true, y_pred, average="macro", zero_division=0)
balanced_acc = balanced_accuracy_score(y_true, y_pred)
mcc = matthews_corrcoef(y_true, y_pred)
kappa = cohen_kappa_score(y_true, y_pred)
cm = confusion_matrix(y_true, y_pred)

print("Accuracy:", accuracy)
print("Macro F1:", macro_f1)
print("Micro F1:", micro_f1)
print("Weighted F1:", weighted_f1)
print("Macro Precision:", macro_precision)
print("Macro Recall:", macro_recall)
print("Balanced Accuracy:", balanced_acc)
print("Matthews Corrcoef:", mcc)
print("Cohen Kappa:", kappa)
print("\nClassification Report:")
print(classification_report(y_true, y_pred, zero_division=0))
print("\nConfusion Matrix:")
print(cm)

result_df.to_csv(f"{model_name}_predictions.csv", index=False)

all_results.append({
    "model_name": model_name,
    "model_id": model_id,
    "accuracy": accuracy,
    "macro_f1": macro_f1,
    "micro_f1": micro_f1,
    "weighted_f1": weighted_f1,
    "macro_precision": macro_precision,
    "macro_recall": macro_recall,
    "balanced_accuracy": balanced_acc,
    "mcc": mcc,
    "cohen_kappa": kappa
})

all_predictions[model_name] = result_df

```

except Exception as e:

```

print(f"Error while testing {model_name}: {e}")
all_results.append({
    "model_name": model_name,
    "model_id": model_id,
    "accuracy": None,
    "macro_f1": None,
    "micro_f1": None,
    "weighted_f1": None,
    "macro_precision": None,
    "macro_recall": None,
    "balanced_accuracy": None,
    "mcc": None,
    "cohen_kappa": None
})

finally:
    if model is not None:
        del model
    if tokenizer is not None:
        del tokenizer
    clear_memory()

```

Out[15]:

```

=====
Loading model: mistral_7b_instruct_v02 -> mistralai/Mistral-7B-Instruct-v0.3

```

Out[15]:

```

Loading weights:   0%|          | 0/291 [00:00<?, ?it/s]

```

Out[15]:

```

mistral_7b_instruct_v02: 100%|██████████| 4276/4276 [12:20<00:00, 5.78it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
  warnings.warn("y_pred contains classes not in y_true")

```

Out[15]:

```

Accuracy: 0.5853601496725912
Macro F1: 0.47313940681995614
Micro F1: 0.5853601496725912
Weighted F1: 0.5964506041462649
Macro Precision: 0.48103521861073645
Macro Recall: 0.4873675748103696
Balanced Accuracy: 0.5685955039454312
Matthews Corrcoef: 0.4924582175650124
Cohen Kappa: 0.48921715561385615

```

Classification Report:

	precision	recall	f1-score	support
Analyze	0.62	0.52	0.57	619
Apply	0.70	0.57	0.63	1160
Create	0.67	0.54	0.60	591
Evaluate	0.48	0.70	0.57	530
Invalid	0.00	0.00	0.00	0
Remember	0.22	0.45	0.30	237

Understand	0.68	0.64	0.66	1139
accuracy			0.59	4276
macro avg	0.48	0.49	0.47	4276
weighted avg	0.62	0.59	0.60	4276

Confusion Matrix:

```
[[324  56  23 135   0  15  66]
 [ 54 658  91 136   1  94 126]
 [ 35 128 317  33  12  29  37]
 [ 40  33  24 370   1   7  55]
 [  0   0   0   0   0   0   0]
 [ 33  16   3  17   0 106  62]
 [ 35  50  16  85   0 225 728]]
```

=====

Loading model: llama3_8b_instruct -> meta-llama/Llama-3.1-8B-Instruct

Out[15]:

config.json: 0%| | 0.00/855 [00:00<?, ?B/s]

Out[15]:

tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]:

tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]:

special_tokens_map.json: 0%| | 0.00/296 [00:00<?, ?B/s]

Out[15]:

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]:

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]:

Fetching 4 files: 0%| | 0/4 [00:00<?, ?it/s]

Out[15]:

Loading weights: 0%| | 0/291 [00:00<?, ?it/s]

Out[15]:

generation_config.json: 0%| | 0.00/184 [00:00<?, ?B/s]

Out[15]:

llama3_8b_instruct: 0%| | 0/4276 [00:00<?, ?it/s]The following generation flags are not valid and may be ignored: ['temperature', 'top_p']. Set `TRANSFORMERS\_VERBOSITY=info` for more details.

llama3_8b_instruct: 100%|██████████| 4276/4276 [39:31<00:00, 1.80it/s]

/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning: y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Out[15]:

Accuracy: 0.514265668849392
Macro F1: 0.3941772253563309

Micro F1: 0.514265668849392
Weighted F1: 0.5322228820496367
Macro Precision: 0.43725183844614435
Macro Recall: 0.3808039504390978
Balanced Accuracy: 0.4442712755122808
Matthews Corrcoeff: 0.41315758632719674
Cohen Kappa: 0.4049637651907986

Classification Report:

	precision	recall	f1-score	support
Analyze	0.54	0.42	0.47	619
Apply	0.76	0.50	0.60	1160
Create	0.56	0.31	0.40	591
Evaluate	0.47	0.66	0.55	530
Invalid	0.00	0.00	0.00	0
Remember	0.14	0.07	0.09	237
Understand	0.59	0.72	0.65	1139
accuracy			0.51	4276
macro avg	0.44	0.38	0.39	4276
weighted avg	0.59	0.51	0.53	4276

Confusion Matrix:

```
[[258  37  12 155  59   6  92]
 [ 95 579  79  70 145  38 154]
 [ 61  84 181  80  52  14 119]
 [ 27  12  20 350  55   3  63]
 [  0   0   0   0   0   0   0]
 [ 14   8   5  11  51  16 132]
 [ 22  45  27  71 119  40 815]]
```

=====

Loading model: phi35_mini_4k_instruct -> microsoft/Phi-3.5-mini-instruct

config.json: 0.00B [00:00, ?B/s]

This model config has set a `rope\_parameters['original\_max\_position\_embeddings']` field, to be used together with `max\_position\_embeddings` to determine a scaling factor. Please set the `factor` field of `rope\_parameters` with this ratio instead -- we recommend the use of this field over `original\_max\_position\_embeddings`, as it is compatible with most model architectures.

tokenizer_config.json: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

tokenizer.model: 0% | 0.00/500k [00:00<?, ?B/s]

added_tokens.json: 0% | 0.00/306 [00:00<?, ?B/s]

special_tokens_map.json: 0%| | 0.00/665 [00:00<?, ?B/s]

Out[15]:

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]:

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]:

Fetching 2 files: 0%| | 0/2 [00:00<?, ?it/s]

Out[15]:

Loading weights: 0%| | 0/195 [00:00<?, ?it/s]

Out[15]:

generation_config.json: 0%| | 0.00/195 [00:00<?, ?B/s]

Out[15]:

phi35_mini_4k_instruct: 100%|██████████| 4276/4276 [23:59<00:00, 2.97it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Out[15]:

Accuracy: 0.5519176800748363
Macro F1: 0.4304912635140586
Micro F1: 0.5519176800748363
Weighted F1: 0.5646765191426816
Macro Precision: 0.45372029874882525
Macro Recall: 0.4342456056372678
Balanced Accuracy: 0.5066198732434791
Matthews Corrcoeff: 0.4571990979495405
Cohen Kappa: 0.44981391769023293

Classification Report:

	precision	recall	f1-score	support
Analyze	0.39	0.71	0.50	619
Apply	0.75	0.49	0.59	1160
Create	0.66	0.47	0.55	591
Evaluate	0.54	0.56	0.55	530
Invalid	0.00	0.00	0.00	0
Remember	0.14	0.16	0.15	237
Understand	0.70	0.65	0.67	1139
accuracy			0.55	4276
macro avg	0.45	0.43	0.43	4276
weighted avg	0.61	0.55	0.56	4276

Confusion Matrix:

```
[[442  32  11  89  12   5  28]
 [201 569  95  62  10  76 147]
 [ 64 104 277  30  47  19  50]
 [161  22  21 299   1   2  24]
 [  0   0   0   0   0   0   0]
```

```
[ 97  8  2 20  3 37 70]
[178 24 11 54  7 129 736]]
```

```
=====
Loading model: qwen2_7b_instruct -> Qwen/Qwen2-7B-Instruct
```

Out[15]:

```
config.json:  0%|          | 0.00/663 [00:00<?, ?B/s]
```

Out[15]:

```
tokenizer_config.json: 0.00B [00:00, ?B/s]
```

Out[15]:

```
vocab.json: 0.00B [00:00, ?B/s]
```

Out[15]:

```
merges.txt: 0.00B [00:00, ?B/s]
```

Out[15]:

```
tokenizer.json: 0.00B [00:00, ?B/s]
```

Out[15]:

```
model.safetensors.index.json: 0.00B [00:00, ?B/s]
```

Out[15]:

```
Downloading (incomplete total...): 0.00B [00:00, ?B/s]
```

Out[15]:

```
Fetching 4 files:  0%|          | 0/4 [00:00<?, ?it/s]
```

Out[15]:

```
Loading weights:  0%|          | 0/339 [00:00<?, ?it/s]
```

Out[15]:

```
generation_config.json:  0%|          | 0.00/243 [00:00<?, ?B/s]
```

Out[15]:

```
qwen2_7b_instruct:  0%|          | 0/4276 [00:00<?, ?it/s]The following generation flags are
not valid and may be ignored: ['temperature', 'top_p', 'top_k']. Set
`TRANSFORMERS_VERBOSITY=info` for more details.
```

```
qwen2_7b_instruct: 100%|██████████| 4276/4276 [09:01<00:00, 7.90it/s]
```

```
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
```

```
warnings.warn("y_pred contains classes not in y_true")
```

Out[15]:

```
Accuracy: 0.5596351730589336
```

```
Macro F1: 0.432377046968275
```

```
Micro F1: 0.5596351730589336
```

```
Weighted F1: 0.5595023436054753
```

```
Macro Precision: 0.5197260499572521
```

```
Macro Recall: 0.4364262442216226
```

```
Balanced Accuracy: 0.509163951591893
```

```
Matthews Corrcoeff: 0.4642673719786513
```

```
Cohen Kappa: 0.451500743484598
```

```
Classification Report:
```

```
precision    recall  f1-score   support
```

Analyze	0.35	0.68	0.46	619
Apply	0.60	0.69	0.64	1160
Create	0.80	0.28	0.42	591
Evaluate	0.47	0.67	0.55	530
Invalid	0.00	0.00	0.00	0
Remember	0.54	0.20	0.29	237
Understand	0.88	0.53	0.66	1139
accuracy			0.56	4276
macro avg	0.52	0.44	0.43	4276
weighted avg	0.65	0.56	0.56	4276

Confusion Matrix:

```
[[424  75   9 109   0   0   2]
 [186 797  21 112   0  11  33]
 [ 62 235 168 114   4   1   7]
 [111  44  10 357   0   1   7]
 [  0   0   0   0   0   0   0]
 [107  42   1   7   0  47  33]
 [316 126   2  67   1  27 600]]
```

=====

Loading model: llama31_8b_instruct -> meta-llama/Llama-3.1-8B-Instruct

Loading weights: 0%| | 0/291 [00:00<?, ?it/s]

llama31_8b_instruct: 100%|██████████| 4276/4276 [39:37<00:00, 1.80it/s]

/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:

y_pred contains classes not in y_true

warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.514265668849392

Macro F1: 0.3941772253563309

Micro F1: 0.514265668849392

Weighted F1: 0.5322228820496367

Macro Precision: 0.43725183844614435

Macro Recall: 0.3808039504390978

Balanced Accuracy: 0.4442712755122808

Matthews Corrcoef: 0.41315758632719674

Cohen Kappa: 0.4049637651907986

Classification Report:

	precision	recall	f1-score	support
Analyze	0.54	0.42	0.47	619
Apply	0.76	0.50	0.60	1160
Create	0.56	0.31	0.40	591
Evaluate	0.47	0.66	0.55	530
Invalid	0.00	0.00	0.00	0
Remember	0.14	0.07	0.09	237
Understand	0.59	0.72	0.65	1139

accuracy			0.51	4276
macro avg	0.44	0.38	0.39	4276
weighted avg	0.59	0.51	0.53	4276

Confusion Matrix:

```
[[258  37  12 155  59   6  92]
 [ 95 579  79  70 145  38 154]
 [ 61  84 181  80  52  14 119]
 [ 27  12  20 350  55   3  63]
 [  0   0   0   0   0   0   0]
 [ 14   8   5  11  51  16 132]
 [ 22  45  27  71 119  40 815]]
```

=====

Out[15]: Loading model: qwen25_7b_instruct -> Qwen/Qwen2.5-7B-Instruct

Out[15]: config.json: 0%| | 0.00/663 [00:00<?, ?B/s]

Out[15]: tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]: vocab.json: 0.00B [00:00, ?B/s]

Out[15]: merges.txt: 0.00B [00:00, ?B/s]

Out[15]: tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]: model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]: Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]: Fetching 4 files: 0%| | 0/4 [00:00<?, ?it/s]

Out[15]: Loading weights: 0%| | 0/339 [00:00<?, ?it/s]

Out[15]: generation_config.json: 0%| | 0.00/243 [00:00<?, ?B/s]

Out[15]: qwen25_7b_instruct: 100%|██████████| 4276/4276 [30:36<00:00, 2.33it/s]

Accuracy: 0.6216089803554724
Macro F1: 0.579021082511993
Micro F1: 0.6216089803554724
Weighted F1: 0.6243433909753118
Macro Precision: 0.5795475780904615
Macro Recall: 0.5890816950971738

Balanced Accuracy: 0.5890816950971738
Matthews Corrcoeff: 0.530001048316422
Cohen Kappa: 0.5286802263184662

Classification Report:

	precision	recall	f1-score	support
Analyze	0.56	0.56	0.56	619
Apply	0.66	0.65	0.65	1160
Create	0.64	0.50	0.56	591
Evaluate	0.52	0.72	0.60	530
Remember	0.33	0.42	0.37	237
Understand	0.76	0.69	0.73	1139
accuracy			0.62	4276
macro avg	0.58	0.59	0.58	4276
weighted avg	0.63	0.62	0.62	4276

Confusion Matrix:

```
[[347  72  30 146   3  21]
 [ 83 749 100  85  45  98]
 [ 44 169 294  29   8  47]
 [ 50  36  24 380   3  37]
 [ 39  34   2  19 100  43]
 [ 57  74   8  69 143 788]]
```

Loading model: qwen25_3b_instruct -> Qwen/Qwen2.5-3B-Instruct

config.json: 0%| | 0.00/661 [00:00<?, ?B/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

vocab.json: 0.00B [00:00, ?B/s]

merges.txt: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Fetching 2 files: 0%| | 0/2 [00:00<?, ?it/s]

Loading weights: 0%| | 0/434 [00:00<?, ?it/s]

generation_config.json: 0%| | 0.00/242 [00:00<?, ?B/s]

qwen25_3b_instruct: 100%|██████████| 4276/4276 [38:37<00:00, 1.84it/s]

Out[15]: /usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning: y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.5610383536014967
Macro F1: 0.40480002139696497
Micro F1: 0.5610383536014967
Weighted F1: 0.5441806616778321
Macro Precision: 0.41510294680763604
Macro Recall: 0.4158544777390048
Balanced Accuracy: 0.48516355736217226
Matthews Corrcoeff: 0.4520020276867105
Cohen Kappa: 0.44590811385195883

Classification Report:

	precision	recall	f1-score	support
Analyze	0.52	0.58	0.55	619
Apply	0.68	0.46	0.55	1160
Create	0.54	0.50	0.52	591
Evaluate	0.46	0.53	0.49	530
Invalid	0.00	0.00	0.00	0
Remember	0.12	0.03	0.05	237
Understand	0.58	0.81	0.68	1139
accuracy			0.56	4276
macro avg	0.42	0.42	0.40	4276
weighted avg	0.55	0.56	0.54	4276

Confusion Matrix:

```
[[362  36  24  82   1   2 112]
 [103 532 160 125   0  18 222]
 [ 49 118 294  49   0  16  65]
 [ 93  19  36 281   1   1  99]
 [  0   0   0   0   0   0   0]
 [ 29  23   2  17   0   7 159]
 [ 62  54  30  58   0  12 923]]
```

=====
Loading model: qwen25_1_5b_instruct -> Qwen/Qwen2.5-1.5B-Instruct

config.json: 0%| | 0.00/660 [00:00<?, ?B/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

vocab.json: 0.00B [00:00, ?B/s]

```
Out[15]: merges.txt: 0.00B [00:00, ?B/s]

Out[15]: tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]: model.safetensors: 0%|          | 0.00/3.09G [00:00<?, ?B/s]

Out[15]: Loading weights: 0%|          | 0/338 [00:00<?, ?it/s]

generation_config.json: 0%|          | 0.00/242 [00:00<?, ?B/s]

Out[15]: qwen25_1_5b_instruct: 100%|██████████| 4276/4276 [12:21<00:00, 5.77it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
  warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.11786716557530402
Macro F1: 0.12415755090651939
Micro F1: 0.11786716557530402
Weighted F1: 0.1849113237371596
Macro Precision: 0.470877846401237
Macro Recall: 0.08016720548250937
Balanced Accuracy: 0.09352840639626094
Matthews Corrcoeff: 0.1578637360623187
Cohen Kappa: 0.08269145055072669

Classification Report:
              precision    recall  f1-score   support

   Analyze         0.42         0.20         0.27         619
     Apply         0.74         0.07         0.12        1160
    Create         0.53         0.03         0.05         591
  Evaluate         0.76         0.02         0.05         530
   Invalid         0.00         0.00         0.00           0
  Remember         0.00         0.00         0.00         237
 Understand         0.84         0.24         0.37        1139

 accuracy                   0.12         4276
 macro avg                 0.47         0.08         0.12         4276
weighted avg                 0.65         0.12         0.18         4276

Confusion Matrix:
[[ 126   3   2   4 480   0   4]
 [  28  76   9   0 1026   0  21]
 [   9  13  17   0  542   0  10]
 [  93   2   2  13  415   0   5]
 [   0   0   0   0   0   0   0]
 [  14   1   0   0  210   0  12]
 [  28   8   2   0  828   1 272]]

=====
```

Out[15]: Loading model: smollm2_1_7b_instruct -> HuggingFaceTB/SmolLM2-1.7B-Instruct

Out[15]: config.json: 0%| | 0.00/908 [00:00<?, ?B/s]

Out[15]: tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]: vocab.json: 0.00B [00:00, ?B/s]

Out[15]: merges.txt: 0.00B [00:00, ?B/s]

Out[15]: special_tokens_map.json: 0%| | 0.00/655 [00:00<?, ?B/s]

Out[15]: tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]: model.safetensors: 0%| | 0.00/3.42G [00:00<?, ?B/s]

Out[15]: Loading weights: 0%| | 0/218 [00:00<?, ?it/s]

generation_config.json: 0%| | 0.00/132 [00:00<?, ?B/s]

Out[15]: smollm2_1_7b_instruct: 100%|██████████| 4276/4276 [05:37<00:00, 12.67it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.4714686623012161
Macro F1: 0.31674141355085006
Micro F1: 0.4714686623012161
Weighted F1: 0.4312863390658519
Macro Precision: 0.47984884889051693
Macro Recall: 0.3326289852442273
Balanced Accuracy: 0.38806714945159854
Matthews Corrcoeff: 0.36770707432499433
Cohen Kappa: 0.3266141732283464

Classification Report:

	precision	recall	f1-score	support
Analyze	0.94	0.12	0.21	619
Apply	0.96	0.27	0.42	1160
Create	0.43	0.56	0.48	591
Evaluate	0.61	0.45	0.52	530
Invalid	0.00	0.00	0.00	0
Remember	0.00	0.00	0.00	237
Understand	0.43	0.93	0.58	1139

accuracy			0.47	4276
macro avg	0.48	0.33	0.32	4276
weighted avg	0.64	0.47	0.43	4276

Confusion Matrix:

```
[[ 73   4  45 113 103   0 281]
 [   4 313 261  13  41   0 528]
 [   0   4 331   5  22   0 229]
 [   1   1  79 238  32   0 179]
 [   0   0   0   0   0   0   0]
Out[15]: [   0   2  13   4   2   0 216]
 [   0   3  49  14  12   0 1061]]
```

=====

Out[15]: Loading model: gemma2_2b_instruct -> google/gemma-2-2b-it

Out[15]: config.json: 0%| | 0.00/838 [00:00<?, ?B/s]

Out[15]: tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]: tokenizer.json: 0%| | 0.00/17.5M [00:00<?, ?B/s]

Out[15]: special_tokens_map.json: 0%| | 0.00/636 [00:00<?, ?B/s]

Out[15]: model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]: Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]: Fetching 2 files: 0%| | 0/2 [00:00<?, ?it/s]

Out[15]: Loading weights: 0%| | 0/288 [00:00<?, ?it/s]

generation_config.json: 0%| | 0.00/187 [00:00<?, ?B/s]

Out[15]: gemma2_2b_instruct: 100%|██████████| 4276/4276 [10:26<00:00, 6.82it/s]
 /usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
 y_pred contains classes not in y_true
 warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.002104770813844715
 Macro F1: 0.004290322985294163
 Micro F1: 0.002104770813844715
 Weighted F1: 0.004174644775425689
 Macro Precision: 0.4693877551020408
 Macro Recall: 0.002182060531985374
 Balanced Accuracy: 0.0025457372873162695

Matthews Corrcoef: 0.0210289307915946
Cohen Kappa: 0.0015714871232503924

Classification Report:

	precision	recall	f1-score	support
Analyze	1.00	0.00	0.00	619
Apply	1.00	0.00	0.01	1160
Create	0.00	0.00	0.00	591
Evaluate	0.00	0.00	0.00	530
Invalid	0.00	0.00	0.00	0
Remember	0.29	0.01	0.02	237
Understand	1.00	0.00	0.01	1139
accuracy			0.00	4276
macro avg	0.47	0.00	0.00	4276
weighted avg	0.70	0.00	0.00	4276

Confusion Matrix:

```
[[ 1  0  0  0 618  0  0]
 [ 0  3  0  0 1156  1  0]
 [ 0  0  0  0 591  0  0]
 [ 0  0  1  0 529  0  0]
 [ 0  0  0  0  0  0  0]
 [ 0  0  0  0 235  2  0]
 [ 0  0  0  0 1132  4  3]]
```

Loading model: gemma2_9b_instruct -> google/gemma-2-9b-it

config.json: 0%| | 0.00/857 [00:00<?, ?B/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

tokenizer.json: 0%| | 0.00/17.5M [00:00<?, ?B/s]

special_tokens_map.json: 0%| | 0.00/636 [00:00<?, ?B/s]

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Fetching 4 files: 0%| | 0/4 [00:00<?, ?it/s]

Loading weights: 0%| | 0/464 [00:00<?, ?it/s]

generation_config.json: 0%| | 0.00/173 [00:00<?, ?B/s]

Out[15]: gemma2_9b_instruct: 100%|██████████| 4276/4276 [26:36<00:00, 2.68it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.6749298409728719
Macro F1: 0.5384685770413793
Micro F1: 0.6749298409728719
Weighted F1: 0.675608868048606
Macro Precision: 0.5555489046051652
Macro Recall: 0.5337499548636402
Balanced Accuracy: 0.6227082806742469
Matthews Corrcoeff: 0.593618766870689
Cohen Kappa: 0.5930778514414159

Classification Report:

	precision	recall	f1-score	support
Analyze	0.61	0.63	0.62	619
Apply	0.71	0.68	0.70	1160
Create	0.63	0.71	0.67	591
Evaluate	0.63	0.68	0.65	530
Invalid	0.00	0.00	0.00	0
Remember	0.54	0.28	0.37	237
Understand	0.77	0.76	0.76	1139
accuracy			0.67	4276
macro avg	0.56	0.53	0.54	4276
weighted avg	0.68	0.67	0.68	4276

Confusion Matrix:

[[387 82 27 93 3 2 25]
[47 791 151 33 11 17 110]
[19 91 422 15 14 6 24]
[56 42 38 359 2 1 32]
[0 0 0 0 0 0 0]
[53 35 4 7 6 67 65]
[74 67 29 62 15 32 860]]

=====
Loading model: phi35_mini_instruct -> microsoft/Phi-4-mini-instruct

config.json: 0.00B [00:00, ?B/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

tokenizer.json: 0%| | 0.00/15.5M [00:00<?, ?B/s]

added_tokens.json: 0%| | 0.00/249 [00:00<?, ?B/s]

special_tokens_map.json: 0%| | 0.00/587 [00:00<?, ?B/s]

Out[15]:

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]:

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]:

Fetching 2 files: 0%| | 0/2 [00:00<?, ?it/s]

Out[15]:

Loading weights: 0%| | 0/194 [00:00<?, ?it/s]

generation_config.json: 0%| | 0.00/168 [00:00<?, ?B/s]

Out[15]:

phi35_mini_instruct: 100%|██████████| 4276/4276 [08:43<00:00, 8.16it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.5640785781103835
Macro F1: 0.4523897703057675
Micro F1: 0.5640785781103835
Weighted F1: 0.5630682670687078
Macro Precision: 0.49366596322456224
Macro Recall: 0.48613657986868347
Balanced Accuracy: 0.5671593431801307
Matthews Corrccoef: 0.4800869169979993
Cohen Kappa: 0.4678474024753603

Classification Report:

	precision	recall	f1-score	support
Analyze	0.78	0.35	0.48	619
Apply	0.84	0.39	0.53	1160
Create	0.46	0.70	0.56	591
Evaluate	0.46	0.73	0.56	530
Invalid	0.00	0.00	0.00	0
Remember	0.26	0.51	0.34	237
Understand	0.66	0.72	0.69	1139
accuracy			0.56	4276
macro avg	0.49	0.49	0.45	4276
weighted avg	0.65	0.56	0.56	4276

Confusion Matrix:

```
[[217  37  64 195   0  30  76]
 [ 24 448 309 103   1 110 165]
 [   6  19 416  42   1  43  64]
 [ 10  13  46 386   0  12  63]
 [   0   0   0   0   0   0   0]
```

Out[15]:

```
[ 10  9 11 22  2 121 62]
[ 12  8 51 91  0 153 824]]
```

```
Out[15]: =====
```

```
Loading model: mistral_nemo_instruct -> mistralai/Mistral-Nemo-Instruct-2407
```

```
Out[15]:
```

```
config.json:  0%|          | 0.00/622 [00:00<?, ?B/s]
```

```
Out[15]:
```

```
tokenizer_config.json: 0.00B [00:00, ?B/s]
```

```
Out[15]:
```

```
tokenizer.json: 0.00B [00:00, ?B/s]
```

```
special_tokens_map.json:  0%|          | 0.00/414 [00:00<?, ?B/s]
```

```
Out[15]:
```

```
The tokenizer you are loading from 'mistralai/Mistral-Nemo-Instruct-2407' with an incorrect
regex pattern: https://huggingface.co/mistralai/Mistral-Small-3.1-24B-Instruct-2503/
discussions/84#69121093e8b480e709447d5e. This will lead to incorrect tokenization. You should
set the `fix_mistral_regex=True` flag when loading this tokenizer to fix this issue.
```

```
Out[15]:
```

```
model.safetensors.index.json: 0.00B [00:00, ?B/s]
```

```
Out[15]:
```

```
Downloading (incomplete total...): 0.00B [00:00, ?B/s]
```

```
Out[15]:
```

```
Fetching 5 files:  0%|          | 0/5 [00:00<?, ?it/s]
```

```
Out[15]:
```

```
Loading weights:  0%|          | 0/363 [00:00<?, ?it/s]
```

```
generation_config.json:  0%|          | 0.00/116 [00:00<?, ?B/s]
```

```
Out[15]:
```

```
mistral_nemo_instruct: 100%|██████████| 4276/4276 [12:28<00:00,  5.71it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
  warnings.warn("y_pred contains classes not in y_true")
```

```
Accuracy: 0.47123479887745556
Macro F1: 0.44454081484131475
Micro F1: 0.47123479887745556
Weighted F1: 0.5689859541960774
Macro Precision: 0.5629313257980844
Macro Recall: 0.38274851439881347
Balanced Accuracy: 0.44653993346528237
Matthews Corrcoeff: 0.4332393507975923
Cohen Kappa: 0.3961846104343507
```

```
Classification Report:
```

```
precision    recall  f1-score   support
```

Analyze	0.75	0.45	0.56	619
Apply	0.86	0.44	0.58	1160
Create	0.65	0.37	0.47	591
Evaluate	0.56	0.67	0.61	530
Invalid	0.00	0.00	0.00	0
Remember	0.22	0.22	0.22	237
Understand	0.89	0.53	0.66	1139
accuracy			0.47	4276
macro avg	0.56	0.38	0.44	4276
weighted avg	0.75	0.47	0.57	4276

Confusion Matrix:

```
[[276  27  10 122 173   3   8]
 [ 26 512  84  39 416  56  27]
 [   3  12 218  12 339   4   3]
 [  21   6  18 357 117   1  10]
 [   0   0   0   0   0   0   0]
 [  15   8   1  17 119  53  24]
 [  26  29   5  90 271 119 599]]
```

```
In [16]: for model_name, df in all_predictions.items():
          print("=" * 80)
          print("Model:", model_name)
          print(df["prediction"].value_counts(dropna=False))
```

```
Out[16]: =====
Model: mistral_7b_instruct_v02
prediction
Understand    1074
Apply          941
Evaluate       776
Analyze        521
Remember       476
Create         474
Invalid        14
Name: count, dtype: int64
=====
Model: llama3_8b_instruct
prediction
Understand    1375
Apply          765
Evaluate       737
Invalid        481
Analyze        477
Create         324
Remember       117
Name: count, dtype: int64
=====
Model: phi35_mini_4k_instruct
prediction
```

Analyze	1143
Understand	1055
Apply	759
Evaluate	554
Create	417
Remember	268
Invalid	80

Name: count, dtype: int64

Model: qwen2_7b_instruct

prediction

Apply	1319
Analyze	1206
Evaluate	766
Understand	682
Create	211
Remember	87
Invalid	5

Name: count, dtype: int64

Model: llama31_8b_instruct

prediction

Understand	1375
Apply	765
Evaluate	737
Invalid	481
Analyze	477
Create	324
Remember	117

Name: count, dtype: int64

Model: qwen25_7b_instruct

prediction

Apply	1134
Understand	1034
Evaluate	728
Analyze	620
Create	458
Remember	302

Name: count, dtype: int64

Model: qwen25_3b_instruct

prediction

Understand	1580
Apply	782
Analyze	698
Evaluate	612
Create	546
Remember	56
Invalid	2

Name: count, dtype: int64

Model: qwen25_1_5b_instruct

prediction

Invalid	3501
---------	------

Understand	324
Analyze	298
Apply	103
Create	32
Evaluate	17
Remember	1

Name: count, dtype: int64

Model: smollm2_1_7b_instruct

prediction

Understand	2494
Create	778
Evaluate	387
Apply	327
Invalid	212
Analyze	78

Name: count, dtype: int64

Model: gemma2_2b_instruct

prediction

Invalid	4261
Remember	7
Apply	3
Understand	3
Create	1
Analyze	1

Name: count, dtype: int64

Model: gemma2_9b_instruct

prediction

Understand	1116
Apply	1108
Create	671
Analyze	636
Evaluate	569
Remember	125
Invalid	51

Name: count, dtype: int64

Model: phi35_mini_instruct

prediction

Understand	1254
Create	897
Evaluate	839
Apply	534
Remember	469
Analyze	279
Invalid	4

Name: count, dtype: int64

Model: mistral_nemo_instruct

prediction

Invalid	1435
Understand	671
Evaluate	637


```
Apply          594
Analyze        367
Create         336
Remember       236
Name: count, dtype: int64
```

In [17]:

```
results_summary_df = pd.DataFrame(all_results)
print(results_summary_df)
results_summary_df.to_csv("all_models_summary.csv", index=False)
```

Out[17]:

	model_name	model_id	accuracy \
0	mistral_7b_instruct_v02	mistralai/Mistral-7B-Instruct-v0.3	0.585360
1	llama3_8b_instruct	meta-llama/Llama-3.1-8B-Instruct	0.514266
2	phi35_mini_4k_instruct	microsoft/Phi-3.5-mini-instruct	0.551918
3	qwen2_7b_instruct	Qwen/Qwen2-7B-Instruct	0.559635
4	llama31_8b_instruct	meta-llama/Llama-3.1-8B-Instruct	0.514266
5	qwen25_7b_instruct	Qwen/Qwen2.5-7B-Instruct	0.621609
6	qwen25_3b_instruct	Qwen/Qwen2.5-3B-Instruct	0.561038
7	qwen25_1_5b_instruct	Qwen/Qwen2.5-1.5B-Instruct	0.117867
8	smollm2_1_7b_instruct	HuggingFaceTB/SmolLM2-1.7B-Instruct	0.471469
9	gemma2_2b_instruct	google/gemma-2-2b-it	0.002105
10	gemma2_9b_instruct	google/gemma-2-9b-it	0.674930
11	phi35_mini_instruct	microsoft/Phi-4-mini-instruct	0.564079
12	mistral_nemo_instruct	mistralai/Mistral-Nemo-Instruct-2407	0.471235

	macro_f1	micro_f1	weighted_f1	macro_precision	macro_recall \
0	0.473139	0.585360	0.596451	0.481035	0.487368
1	0.394177	0.514266	0.532223	0.437252	0.380804
2	0.430491	0.551918	0.564677	0.453720	0.434246
3	0.432377	0.559635	0.559502	0.519726	0.436426
4	0.394177	0.514266	0.532223	0.437252	0.380804
5	0.579021	0.621609	0.624343	0.579548	0.589082
6	0.404800	0.561038	0.544181	0.415103	0.415854
7	0.124158	0.117867	0.184911	0.470878	0.080167
8	0.316741	0.471469	0.431286	0.479849	0.332629
9	0.004290	0.002105	0.004175	0.469388	0.002182
10	0.538469	0.674930	0.675609	0.555549	0.533750
11	0.452390	0.564079	0.563068	0.493666	0.486137
12	0.444541	0.471235	0.568986	0.562931	0.382749

	balanced_accuracy	mcc	cohen_kappa
0	0.568596	0.492458	0.489217
1	0.444271	0.413158	0.404964
2	0.506620	0.457199	0.449814
3	0.509164	0.464267	0.451501
4	0.444271	0.413158	0.404964
5	0.589082	0.530001	0.528680
6	0.485164	0.452002	0.445908
7	0.093528	0.157864	0.082691
8	0.388067	0.367707	0.326614
9	0.002546	0.021029	0.001571
10	0.622708	0.593619	0.593078
11	0.567159	0.480087	0.467847

Model Evaluation Metrics

We compute:

- Accuracy
- Macro F1 Score
- Precision Score
- Recall Score
- Confusion Matrix
- Balanced Accuracy Score
- Mathews Correlation
- Cohen's Kappa Score

In [18]:

```
from sklearn.metrics import (
    accuracy_score,
    f1_score,
    precision_score,
    recall_score,
    confusion_matrix,
    classification_report,
    balanced_accuracy_score,
    matthews_corrcoef,
    cohen_kappa_score
)

# remove rows with missing labels
result_df = result_df.dropna(subset=["label", "prediction"])

# clean labels
y_true = result_df["label"].astype(str).str.strip()
y_pred = result_df["prediction"].astype(str).str.strip()

# basic metrics
accuracy = accuracy_score(y_true, y_pred)

macro_f1 = f1_score(y_true, y_pred, average="macro", zero_division=0)
micro_f1 = f1_score(y_true, y_pred, average="micro", zero_division=0)
weighted_f1 = f1_score(y_true, y_pred, average="weighted", zero_division=0)

macro_precision = precision_score(y_true, y_pred, average="macro", zero_division=0)
macro_recall = recall_score(y_true, y_pred, average="macro", zero_division=0)

balanced_acc = balanced_accuracy_score(y_true, y_pred)

# additional research metrics
mcc = matthews_corrcoef(y_true, y_pred)
kappa = cohen_kappa_score(y_true, y_pred)

# confusion matrix
```

```

cm = confusion_matrix(y_true, y_pred)

print("Accuracy:", accuracy)
print("Macro F1:", macro_f1)
print("Micro F1:", micro_f1)
print("Weighted F1:", weighted_f1)

print("Macro Precision:", macro_precision)
print("Macro Recall:", macro_recall)

print("Balanced Accuracy:", balanced_acc)

print("Matthews Corrcoeff:", mcc)
print("Cohen Kappa:", kappa)

print("\nClassification Report:")
print(classification_report(y_true, y_pred, zero_division=0))

print("\nConfusion Matrix:")
print(cm)

```

Out[18]:

```

Accuracy: 0.47123479887745556
Macro F1: 0.44454081484131475
Micro F1: 0.47123479887745556
Weighted F1: 0.5689859541960774
Macro Precision: 0.5629313257980844
Macro Recall: 0.38274851439881347
Balanced Accuracy: 0.44653993346528237
Matthews Corrcoeff: 0.4332393507975923
Cohen Kappa: 0.3961846104343507

```

Classification Report:

	precision	recall	f1-score	support
Analyze	0.75	0.45	0.56	619
Apply	0.86	0.44	0.58	1160
Create	0.65	0.37	0.47	591
Evaluate	0.56	0.67	0.61	530
Invalid	0.00	0.00	0.00	0
Remember	0.22	0.22	0.22	237
Understand	0.89	0.53	0.66	1139
accuracy			0.47	4276
macro avg	0.56	0.38	0.44	4276
weighted avg	0.75	0.47	0.57	4276

Confusion Matrix:

```

[[276  27  10 122 173   3   8]
 [ 26 512  84  39 416  56  27]
 [   3  12 218  12 339   4   3]
 [  21   6  18 357 117   1  10]
 [   0   0   0   0   0   0   0]
 [  15   8   1  17 119  53  24]

```

```
[ 26  29   5  90 271 119 599]]
```

```
Out[18]: /usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
  warnings.warn("y_pred contains classes not in y_true")
```

Save Predictions

```
In [19]: result_df = test_df.copy()
result_df["model_name"] = model_name
result_df["model_id"] = model_id
result_df["prediction"] = predictions

result_df.to_csv(f"{model_name}_predictions.csv", index=False)

y_true = result_df["label"].astype(str).str.strip()
y_pred = result_df["prediction"].astype(str).str.strip()
```